US 20250285417A1

(19) **United States**
(12) **Patent Application Publication**
MORIN et al.
(10) **Pub. No.: US 2025/0285417 A1**
(43) **Pub. Date: Sep. 11, 2025**

(54) **EYE OPENNESS**

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(21) Appl. No.: **19/218,473**

(22) Filed: **May 26, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/175,577, filed on Feb. 28, 2023, now Pat. No. 12,347,173.

(30) **Foreign Application Priority Data**

Mar. 4, 2022 (SE) 2250299-1

Publication Classification

(51) **Int. Cl.**
G06V 10/774 (2022.01)
G06F 3/01 (2006.01)
G06V 40/18 (2022.01)

(52) **U.S. Cl.**
CPC ***G06V 10/774*** (2022.01); ***G06F 3/013*** (2013.01); ***G06V 40/193*** (2022.01)

(57) **ABSTRACT**

A controller configured to: receive first and second curve data respectively defining first and second curves respectively representative of first and second eyelid edges in an eye image; determine an eye-openness-indicator-line extending from a first intersection point on the first curve to a second intersection point on the second curve by performing an optimisation routine comprising: defining an objective function representative of: an orthogonality of the eye-openness-indicator-line to a first tangent to the first curve at the first intersection point; and an orthogonality of the eye openness indicator line to a second tangent to the second curve at the second intersection point; and adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied; and provide an eye openness value based on a length of the eye-openness-indicator-line.

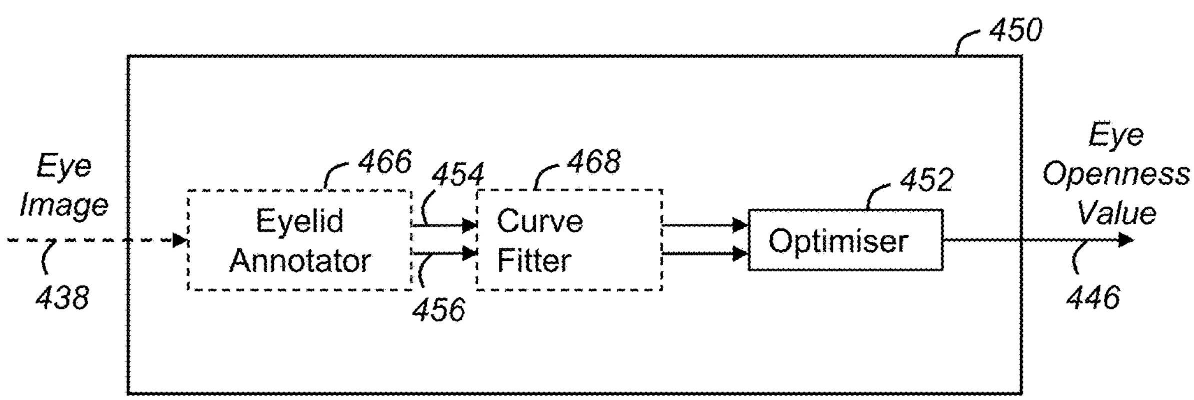


Figure 1

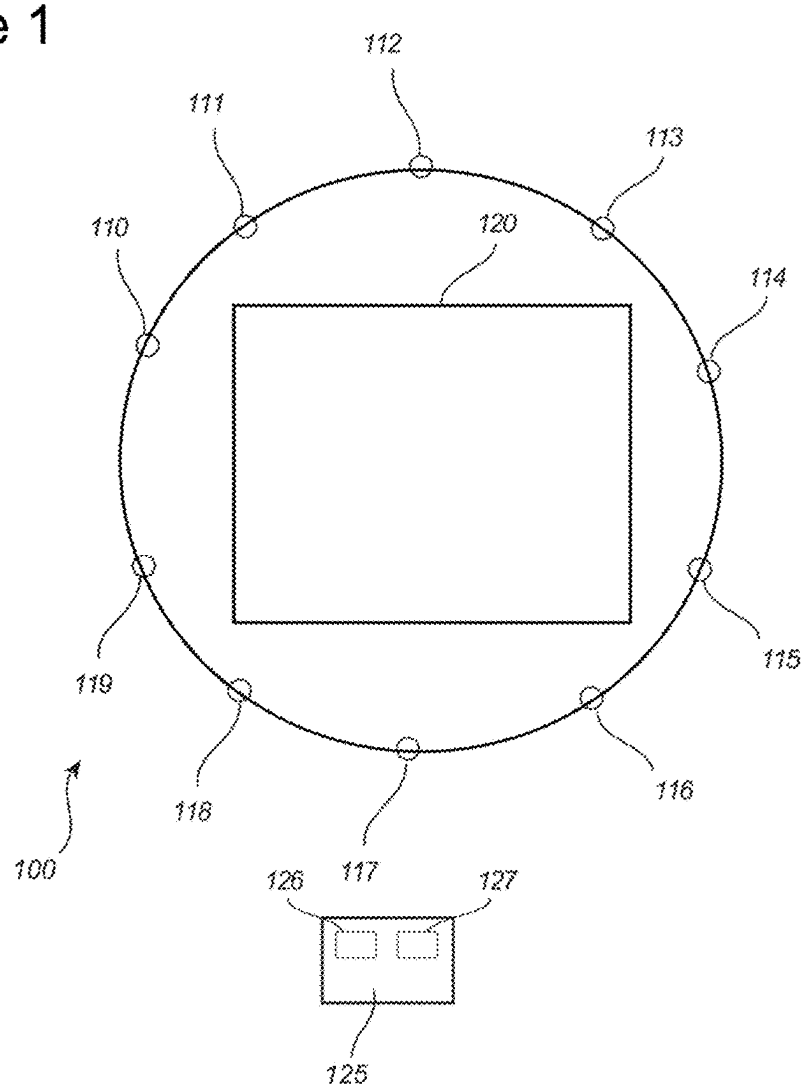


Figure 2

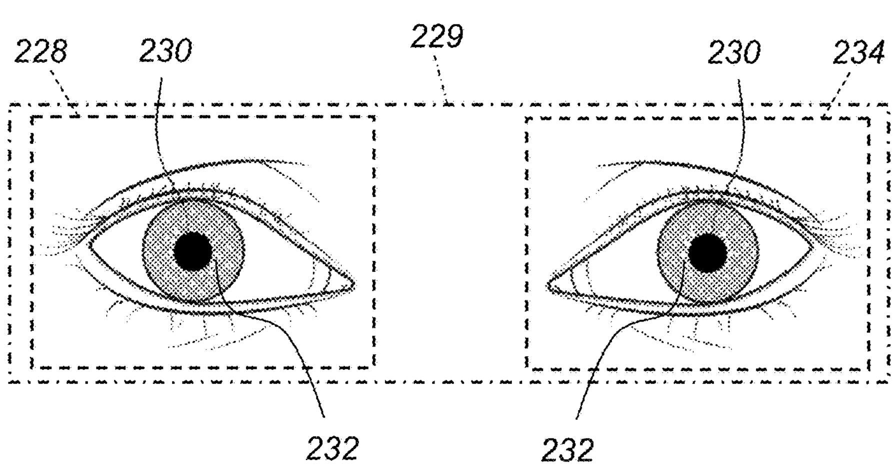


Figure 3

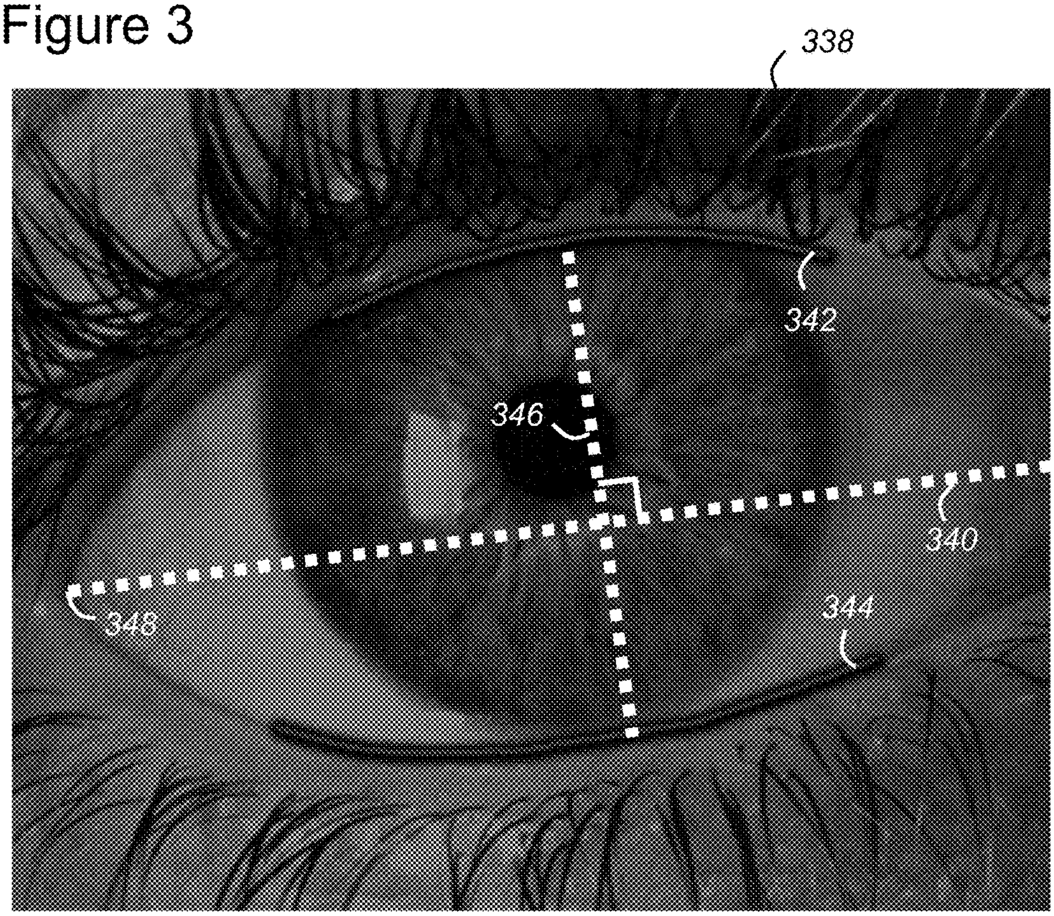


Figure 7

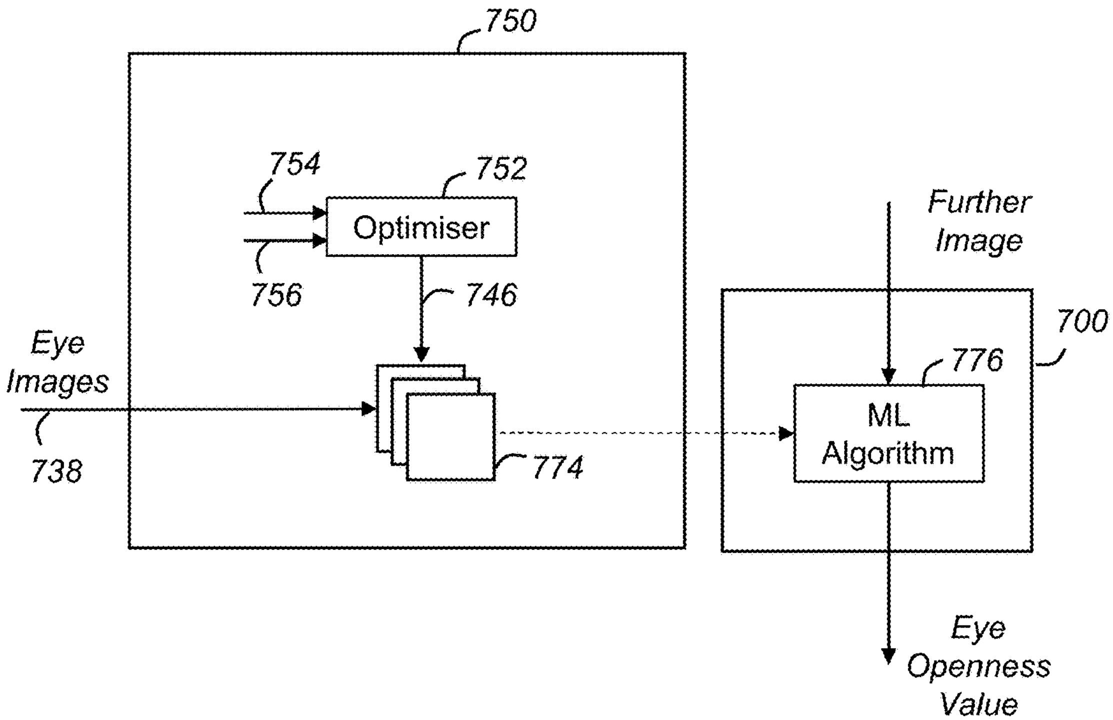


Figure 4

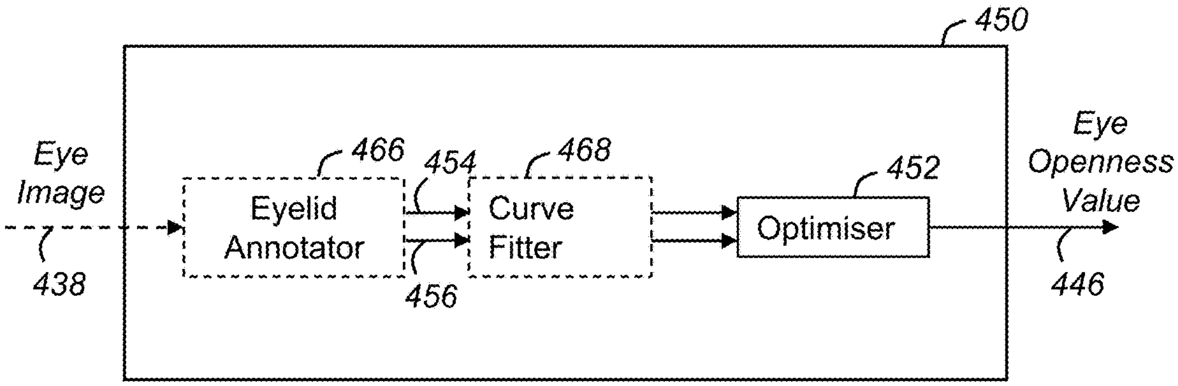
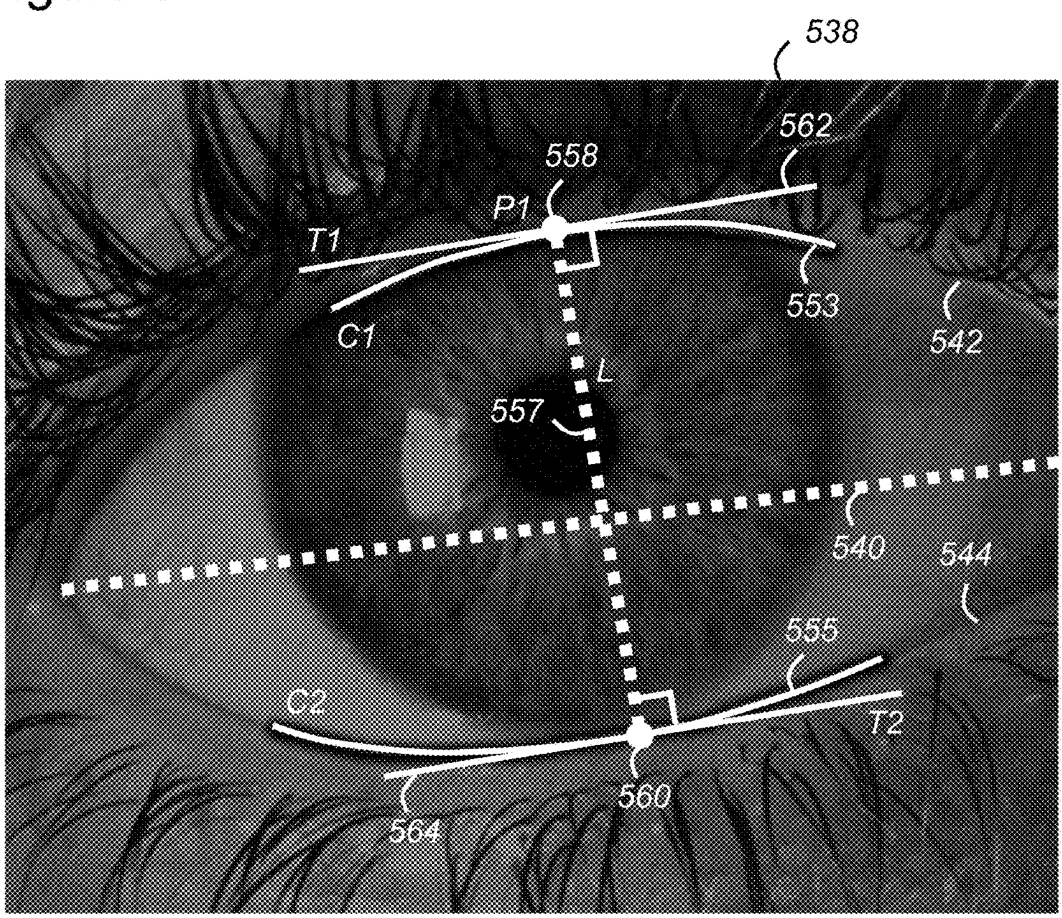


Figure 5



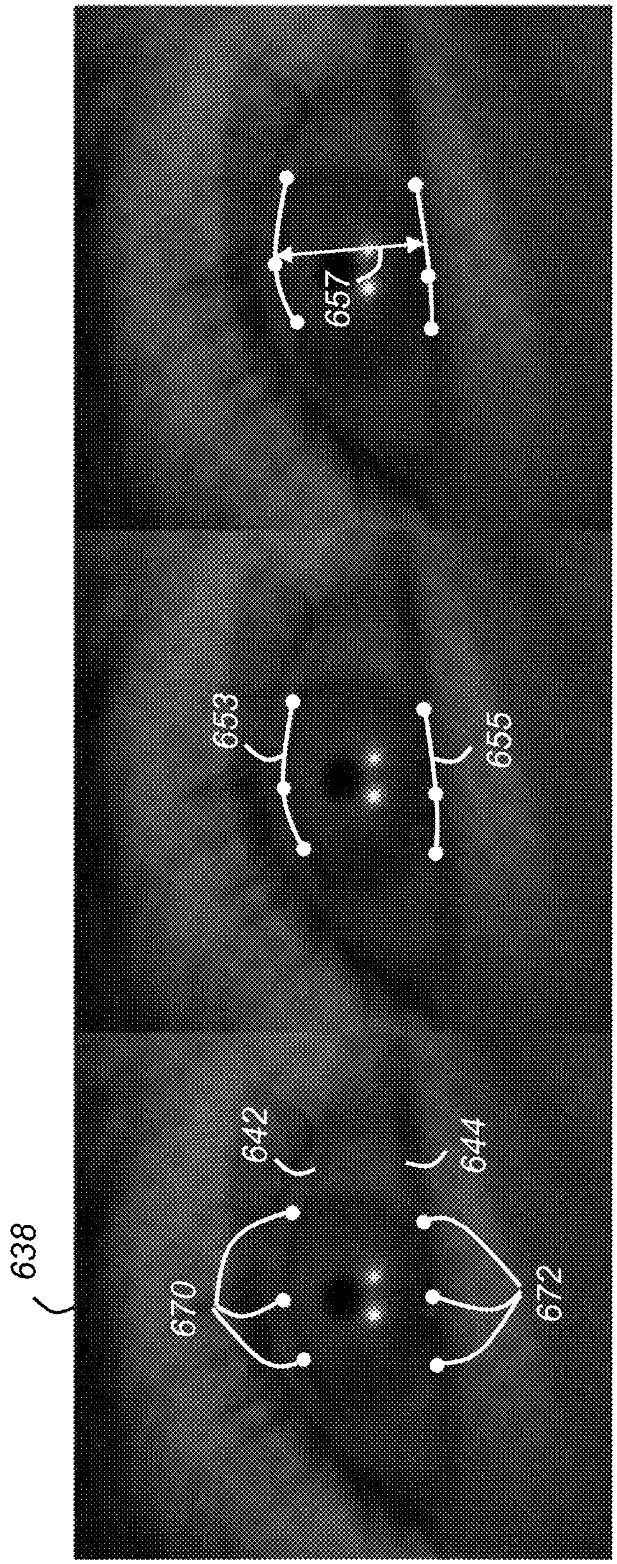


Figure 6

Figure 8

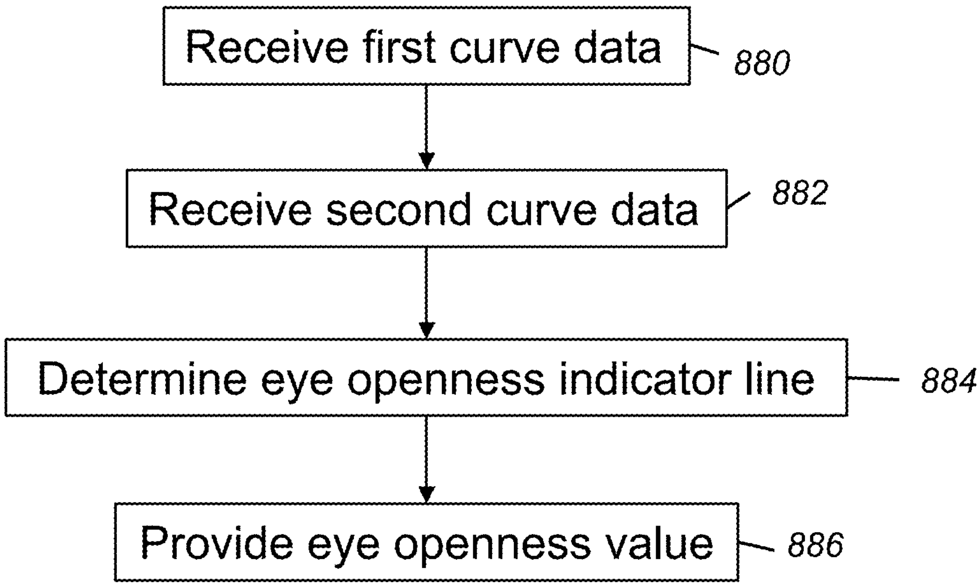
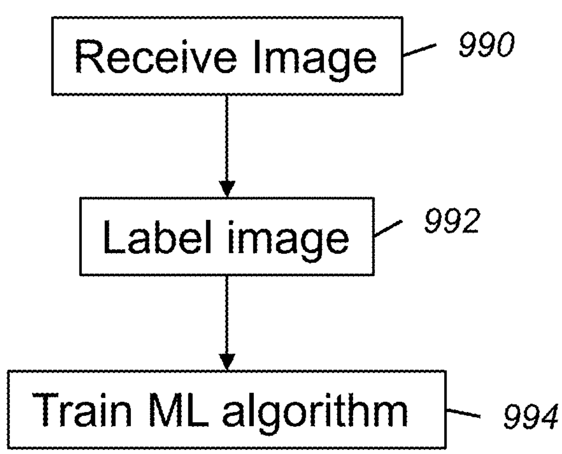


Figure 9



EYE OPENNESS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Swedish patent application No. 2250299-1, filed Mar. 4, 2022, entitled "EYE OPENNESS", and is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure generally relates to the field of eye tracking. In particular, the present disclosure relates to controllers, algorithms, eye tracking systems and methods for determining an eye openness of an eye.

BACKGROUND

[0003] In eye tracking applications, digital images are retrieved of the eyes of a user and the digital images are analysed in order to estimate the gaze direction of the user. The estimation of the gaze direction may be based on computer-based image analysis of features of the imaged eye. One known example method of eye tracking includes the use of infrared light and an image sensor. The infrared light is directed towards the pupil of a user and the reflection of the light is captured by an image sensor.

[0004] Many eye tracking systems estimate gaze direction based on identification of a pupil position together with glints or corneal reflections. However, gaze estimation techniques can depend on eye openness—that is a degree of openness of an eye. For example, blink detection can enable the suspension of gaze tracking during a blink. Therefore, accurately determining eye openness can be important for eye tracking systems and methods.

[0005] Portable or wearable eye tracking devices have also been previously described. One such eye tracking system is described in U.S. Pat. No. 9,041,787 (which is hereby incorporated by reference in its entirety). A wearable eye tracking device is described using illuminators and image sensors for determining gaze direction.

SUMMARY

[0006] According to a first aspect of the present disclosure there is provided a controller configured to:

[0007] receive first curve data defining a first curve representative of a first eyelid edge in an image of an eye;

[0008] receive second curve data defining a second curve representative of a second eyelid edge in the image of the eye;

[0009] determine an eye openness indicator line extending from a first intersection point on the first curve to a second intersection point on the second curve by performing an optimisation routine comprising:

[0010] defining an objective function representative of:

[0011] an orthogonality of the eye openness indicator line to a first tangent to the first curve at the first intersection point; and

[0012] an orthogonality of the eye openness indicator line to a second tangent to the second curve at the second intersection point; and

[0013] adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied; and

[0014] provide an eye openness value based on a length of the eye openness indicator line.

[0015] The objective function may be based on:

[0016] a first angle between the eye openness indicator line and a normal to the tangent to the first curve at the first intersection point; and

[0017] a second angle between the eye openness indicator line and a normal to the tangent to the second curve at the second intersection point.

[0018] The objective function may be based on:

[0019] a dot product of the eye openness indicator line and the tangent to the first curve at the first intersection point; and

[0020] a dot product of the eye openness indicator line and the tangent to the second curve at the second intersection point.

[0021] The objective function may be based on a sum of squares of, or a sum of magnitudes of:

[0022] the dot product of the eye openness indicator line and the tangent to the first curve at the first intersection point; and

[0023] the dot product of the eye openness indicator line and the tangent to the second curve at the second intersection point.

[0024] The objective function may be further based on the length of the eye openness indicator line.

[0025] The optimisation routine may comprise adjusting the value of the first intersection point and the value of the second intersection point to reduce a value of the objective function until the at least one termination condition for the value of the objective function is satisfied.

[0026] The at least one termination condition may comprise at least one of:

[0027] the value of the objective function is less than a threshold value;

[0028] the value of the objective function is greater than a threshold value;

[0029] a variation of a fixed number of successive values of the objective function converge within a convergence threshold; and

[0030] a maximum number of iterations of adjusting the value of the first intersection point and the value of the second intersection point have been performed.

[0031] The controller may be configured to:

[0032] receive the first curve data as first curve data comprising at least three first data points associated with the first eyelid edge;

[0033] receive the second curve data as second curve data comprising at least three second data points associated with the second eyelid edge;

[0034] curve fit the at least three first data points to define the first curve; and

[0035] curve fit the at least three second data points to define the first curve.

[0036] The controller may be configured to curve fit the at least three first data points and the at least three second data points using any of:

[0037] a second order polynomial function;

[0038] a polynomial function of an order higher than second order; an elliptical function;

[0039] a parabolic function;
[0040] a hyperbolic function; and
[0041] a trigonometric function.
[0042] The controller may be configured to:
[0043] receive the image of the eye;
[0044] determine the at least three first data points using image processing; and
[0045] determine the at least three second data points using image processing.
[0046] The controller may be configured to:
[0047] receive the image of the eye;
[0048] label the image of the eye with the eye openness value to provide labelled training data.
[0049] The controller may be configured to:
[0050] receive a plurality of eye images and first and second curve data associated with each eye image;
[0051] determine an eye openness indicator line for each of the plurality of eye images by performing the optimisation routine; and
[0052] label each eye image with an eye openness value based on a length of the respective eye openness indicator line to provide labelled training data.
[0053] The controller may be configured to:
[0054] train a machine learning algorithm to output eye openness values in response to eye image inputs using the labelled training data.
[0055] According to a second aspect of the present disclosure there is provided a machine learning algorithm trained by any controller disclosed herein.
[0056] According to a third aspect of the present disclosure there is provided an eye tracking system comprising:
[0057] a memory storing any machine learning algorithm disclosed herein; and
[0058] a processor configured to:
[0059] receive a further image of an eye; and
[0060] output an eye openness value by processing the image of the eye using the machine learning algorithm.
[0061] According to a fourth aspect of the present disclosure there is provided an eye tracking system comprising any controller disclosed herein controller.
[0062] According to a fifth aspect of the present disclosure there is provided a method for determining an eye openness value of an image of an eye, the method comprising:
[0063] receiving first curve data defining a first curve representative of a first eyelid edge in an image of an eye;
[0064] receiving second curve data defining a second curve representative of a second eyelid edge in the image of the eye;
[0065] determining an eye openness indicator line extending from a first intersection point on the first curve to a second intersection point on the second curve by performing an optimisation routine comprising:
[0066] defining an objective function representative of:
[0067] an orthogonality of the eye openness indicator line and a tangent to the first curve at the first intersection point; and
[0068] an orthogonality of the eye openness indicator line and a tangent to the second curve at the second intersection point; and
[0069] adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied; and
[0070] providing an eye openness value based on a length of the eye openness indicator line.
[0071] The method of claim **15** may further comprise:
[0072] receiving the image of the eye;
[0073] labelling the image of the eye with the eye openness value to provide labelled training data; and
[0074] training a machine learning algorithm to output eye openness values in response to eye image inputs using the labelled training data.
[0075] According to a sixth aspect of the present disclosure there is provided a machine learning algorithm obtainable by any method disclosed herein.
[0076] According to a seventh aspect of the present disclosure there is provided a head-mounted device comprising any eye tracking system disclosed herein.
[0077] According to an eight aspect of the present disclosure there is provided one or more non-transitory computer-readable storage media storing computer-executable instructions that, when executed by a computing system, causes the computing system to perform any method disclosed herein.
[0078] According to a ninth aspect of the present disclosure there is provided a controller configured to:
[0079] receive first curve data defining a first curve representative of a first eyelid edge of an image of an eye;
[0080] receive second curve data defining a second curve representative of a second eyelid edge of the image of the eye;
[0081] determine an eye openness indicator line extending from a first point on the first curve to a second point on the second curve by selecting the first point and the second point based on:
[0082] an orthogonality of a resultant first angle between the eye openness indicator line and a tangent to the first curve at the first point; and
[0083] an orthogonality of a resultant second angle between the eye openness indicator line and a tangent to the second curve at the second point; and
[0084] provide an eye openness value based on a length of the eye openness indicator line.
[0085] There may be provided a computer program, which when run on a computer, causes the computer to configure any apparatus, including a circuit, controller or device disclosed herein or perform any method disclosed herein. The computer program may be a software implementation, and the computer may be considered as any appropriate hardware, including a digital signal processor, a microcontroller, and an implementation in read only memory (ROM), erasable programmable read only memory (EPROM) or electronically erasable programmable read only memory (EEPROM), as non-limiting examples. The software may be an assembly program.
[0086] The computer program may be provided on a computer readable medium, which may be a physical computer readable medium such as a disc or a memory device, or may be embodied as a transient signal. Such a transient signal may be a network download, including an internet download. There may be provided one or more non-transitory computer-readable storage media storing computer-executable instructions that, when executed by a computing system, causes the computing system to perform any method disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0087] One or more embodiments will now be described by way of example only with reference to the accompanying drawings in which:

[0088] FIG. **1** shows a schematic view of an eye tracking system which may be used to capture a sequence of images that can be used by example embodiments of the present disclosure;

[0089] FIG. **2** shows an example image of a pair of eyes;

[0090] FIG. **3** illustrates an example measure of eye openness;

[0091] FIG. **4** illustrates a controller for providing an eye openness value of an image according to an embodiment of the disclosure;

[0092] FIG. **5** illustrates the geometrical relationships used by the controller of FIG. **4**;

[0093] FIG. **6** schematically illustrates operational steps performed by a controller according to an embodiment of the disclosure;

[0094] FIG. **7** illustrates another controller for providing an eye openness value and an eye tracking system according to embodiments of the present disclosure;

[0095] FIG. **8** illustrates schematically a method according to an embodiment of the present disclosure; and

[0096] FIG. **9** illustrates schematically a further method according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0097] FIG. **1** shows a simplified view of an eye tracking system **100** (which may also be referred to as a gaze tracking system) in a head-mounted device in the form of a virtual or augmented reality (VR or AR) device or VR or AR glasses or anything related, such as extended reality (XR) or mixed reality (MR) headsets. The system **100** comprises an image sensor **120** (e.g. a camera) for capturing images of the eyes of the user. The system may optionally include one or more illuminators **110-119** for illuminating the eyes of a user, which may for example be light emitting diodes emitting light in the infrared frequency band, or in the near infrared frequency band and which may be physically arranged in a variety of configurations. The image sensor **120** may for example be an image sensor of any type, such as a complementary metal oxide semiconductor (CMOS) image sensor or a charged coupled device (CCD) image sensor. The image sensor may consist of an integrated circuit containing an array of pixel sensors, each pixel containing a photodetector and an active amplifier. The image sensor may be capable of converting light into digital signals. In one or more examples, it could be an Infrared image sensor or IR image sensor, an RGB sensor, an RGBW sensor or an RGB or RGBW sensor with IR filter.

[0098] The eye tracking system **100** may comprise circuitry or one or more controllers **125**, for example including a receiver **126** and processing circuitry **127**, for receiving and processing the images captured by the image sensor **120**. The circuitry **125** may for example be connected to the image sensor **120** and the optional one or more illuminators **110-119** via a wired or a wireless connection and be co-located with the image sensor **120** and the one or more illuminators **110-119** or located at a distance, e.g. in a different device. In another example, the circuitry **125** may be provided in one or more stacked layers below the light sensitive surface of the light sensor **120**.

[0099] The eye tracking system **100** may include a display (not shown) for presenting information and/or visual stimuli to the user. The display may comprise a VR display which presents imagery and substantially blocks the user's view of the real-world or an AR display which presents imagery that is to be perceived as overlaid over the user's view of the real-world.

[0100] The location of the image sensor **120** for one eye in such a system **100** is generally away from the line of sight for the user in order not to obscure the display for that eye. This configuration may be, for example, enabled by means of so-called hot mirrors which reflect a portion of the light and allows the rest of the light to pass, e.g. infrared light is reflected, and visible light is allowed to pass.

[0101] While in the above example the images of the user's eye are captured by a head-mounted image sensor **120**, in other examples the images may be captured by an image sensor that is not head-mounted. Such a non-head-mounted system may be referred to as a remote system.

[0102] In an eye tracking system, a gaze signal can be computed for each eye of the user (left and right). The quality of these gaze signals can be reduced by disturbances in the input images (such as image noise) and by incorrect algorithm behaviour (such as incorrect predictions). A goal of the eye tracking system is to deliver a gaze signal that is as good as possible, both in terms of accuracy (bias error) and precision (variance error). For many applications it can be sufficient to deliver only one gaze signal per time instance, rather than both the gaze of the left and right eyes individually. Further, the combined gaze signal can be provided in combination with the left and right signals. Such a gaze signal can be referred to as a combined gaze signal.

[0103] FIG. **2** shows a simplified example of an image **229** of a pair of eyes, captured by an eye tracking system such as the system of FIG. **1**. The image **229** can be considered as including a right-eye-image **228**, of a person's right eye, and a left-eye-image **234**, of the person's left eye. In this example the right-eye-image **228** and the left-eye-image **234** are both parts of a larger image of both person's eyes. In other examples, separate image sensors may be used to acquire the right-eye-image **228** and the left-eye-image **234**.

[0104] The system may employ image processing (such as digital image processing) for extracting features in the image. The system may for example identify a position of the pupil **230** in the one or more images captured by the image sensor. The system may determine the position of the pupil **230** using a pupil detection process. The system may also identify corneal reflections **232** located in close proximity to the pupil **230**. The system may estimate a corneal centre or eyeball centre based on the corneal reflections **232**. For example, the system may match each of the individual corneal reflections **232** for each eye with a corresponding illuminator and determine the corneal centre of each eye based on the matching.

[0105] Eye openness is a bio-metric signal used in a large range of applications of eye tracking systems, for example the eye tracking system of FIG. **1**. Example applications include behavioural and cognitive research, automotive driver monitoring systems and gaze signal filtering and/or validation.

[0106] FIG. **3** illustrates an example measure of eye openness **346** for an image of an eye **338**. In this example, eye openness **346** can be defined as a maximum distance between an upper eye lid **342** and a lower eyelid **344** in a direction orthogonal to a horizontal eye-axis **340** connecting a first corner of the eye **348** to a second corner of the eye (not shown). In this context, "eyelid" is the line segment that constitutes the edge visually separating an eyeball and eyelid skin.

[0107] Eye tracking systems can use an automated process to determine eye openness, for example, a classic rule-based algorithm or a machine learning (ML) algorithm. An ML algorithm can be trained using eye images labelled (or paired) with ground truth values of eye openness. A rule-based algorithm for determining eye openness may also require ground truth values in order to measure the accuracy of the determination.

[0108] For determining the ground truth value of eye openness, acquiring the actual 3D distance (a value represented in millimeters) between eyelids is challenging. Obtaining an accurate 3D distance can require a non-intrusive device that can measure the 3D distance with precision and accuracy. An alternative approach is to determine eye openness in the (2D) image domain in which the ground truth eye openness can be annotated. For example, an image of an eye **338** can be labelled with an eye openness value **346** measured according to the example of FIG. **3** and a plurality of such labelled images can be used as training data for the ML algorithm. The image domain approach works particularly well for a small or zero perspective between the camera and the eye. However, larger perspectives can be corrected using camera parameters, a camera-eye distance, a size of the eye in the image etc.

[0109] Image labelling in the image domain is typically performed manually and relies on subjective human judgement in relation to the eyelid edges **342**, **344** and the maximum vertical distance **346**. Annotating the largest orthogonal distance between upper and lower eyelids **342**, **344** can be difficult and inconsistent and relies heavily on the subjective assessment of the person annotating the image. The annotation requires the annotator to visually find a pair of points where the eye is the most open, which is time consuming and difficult. As a result, the image labelling process is time consuming and the resulting training data and ML algorithm can lack accuracy and repeatability.

[0110] The disclosed apparatus and methods can determine eye openness for eye images. The apparatus and methods can infer image domain eye openness from an image of an eye or from a set of landmark points along the upper and lower eyelid edges **342**, **344** of the image. The apparatus and methods can advantageously provide accurate and repeatable values of eye openness for eye images. The eye openness values may be used for labelling training data for an eye openness ML algorithm and/or for determining eye openness values in an eye tracking system.

[0111] FIG. **4** illustrates a controller **450** for providing an eye openness value **446** of an image of an eye **438**, according to an embodiment of the present disclosure. The controller **450** is described with reference to the image of the eye **538** in FIG. **5**.

[0112] In this example, the controller **450** comprises an optimiser **452** configured to receive first curve data **454** and second curve data **456** (optionally via a curve fitter **468**). The first and second curve data **454**, **456** respectively define a first curve, C**1**, **553** and second curve, C**2**, **555** each representative of a respective first (upper) eyelid edge **542** and second (lower) eyelid edge **544** in/of the image of the eye **438**, **538**. The optimiser **452** is configured to determine an eye openness indicator line, L, **557** extending from a first intersection point, P**1**, **558** on the first curve **553** to a second intersection point, P**2**, **560** on the second curve **555** by performing an optimisation routine. The optimisation routine comprises first defining an objective function representative of: an orthogonality of the eye openness indicator line **557** to a first tangent, T**1**, **562** of the first curve **553** at the first intersection point **558**; and an orthogonality of the eye openness indicator line **557** to a second tangent, **72**, **564** of the second curve **555** at the second intersection point **560**. The optimisation routine further comprises adjusting a value of the first intersection point **558** and the second intersection point **560** until at least one termination condition for a value of the objective function is satisfied. For example, the first and second intersection points **558**, **560** may be adjusted until a minimum value of the objective function is obtained such that the two angles between eye openness indicator line **557** and the first and second tangents **562**, **564** are as close to 90 degrees as possible. Following determination of the eye openness indicator line **557** using the optimisation routine, the controller **450** is configured to provide or output the eye openness value **446** based on a length of the eye openness indicator line **557**.

[0113] The optimisation routine can find the pair of first and second intersection points, P**1**, P**2** on the first and second curves, C**1**, C**2** such that the eye openness indicator line extending between them, L(P**1**, P**2**), is orthogonal (or as close to orthogonal as possible) to the first and second tangents, T**1**, T**2**, where T**1** equals the tangent to C**1** at P**1** and T**2** equals the tangent to C**2** at P**2**.

[0114] The optimisation routine may comprise:

[0115] (i) Defining the objective function as a residual function R(P**1**, P**2**) based on the angles between the eye openness indicator line, L(P**1**,P**2**), and the first and second tangents, T**1**, T**2**. For example, the objective function may be based on: a dot product of the eye openness indicator line, L(P**1**,P**2**), and the first tangent T**1**; and a dot product of the eye openness indicator line, L(P**1**,P**2**), and the second tangent, T**2**. For example, the residual function may be based on (or equal to) the sum of the squares of the two dot products:

$$R(P1, P2) = f\left((T1 \cdot L)^2 + (T2 \cdot L)^2\right)$$

[0116] When the first and second tangents T**1**, T**2** are both orthogonal to the eye openness indicator line, L(P**1**, P**2**), the residual function will be zero. In some examples, the sum of the squares of the dot products may comprise a weighted sum. In some examples, each dot product may be normalised by dividing by the magnitudes of the eye openness indicator line and the respective tangent. In some examples, the residual function may be based on or equal to a sum of the magnitudes of the two dot products.

[0117] A further example residual function may be based on: a first angle, α**1**, between the eye openness indicator line, L(P**1**,P**2**), and a normal to the first tangent, T**1**; and a second angle, α**2**, between the eye openness indicator line, L(P**1**,P**2**), and a normal to the second tangent, T**2**. In some examples, the residual function may be based on (or equal to) the sum of the magnitudes of the two angles:

$$R(P1, P2) = f(|\alpha 1| + |\alpha 2|)$$

[0118] When the first and second tangents T**1**, T**2** are both orthogonal to the eye openness indicator line, L(P**1**, P**2**), the residual function will be zero.

[0119] (ii) Using a numerical optimiser to find values of the first and second intersection points, P**1**, P**2** that minimize the objective function R(P**1**,P**2**). The numerical optimiser may comprise known optimisation algorithms, for example, a Nelder-Mead method, a hill climb/descent routine or any other known optimisation method. The optimisation method may comprise iteratively adjusting the values of the first and second intersection points, P**1**, P**2**, and computing the resultant objective function to determine a minimum of the objective function. The adjusted values of the first and second intersection points, P**1**(n), P**2**(n), for a particular iteration may be selected based on the first and second intersection points, P**1**(n−1, n−2, . . .), P**2**(n−1, n−2, . . .) and the resulting objective function values R(P**1** (n−1, n−2, . . .), P**2**(n−1, n−2, . . .)) from one or more previous iterations.

[0120] Under typical conditions (where the area between the first and second eyelids **542**, **544** is convex and the eyelid curvatures are small enough) the eye openness indicator line segment L(P**1**,P**2**) between those intersection points P**1**, P**2** that minimize the orthogonality-based objective function R will also be a diameter of the largest circular disk that fits between the eyelids. Therefore, the controller **450** is configured to provide the eye openness value **446** based on the length of the eye openess indicator line **557**. For example, the controller **450** may provide the eye openness value **446** based on a magnitude of the length of the eye openness indicator line, |L(P**1**,P**2**)|.

[0121] In some examples, the objective function may be further based on the length of the eye openness indicator line **557**. For example, the sum (or weighted sum) in the above objective functions may comprise an additional term inversely proportional to the length of the eye openness indicator line **557**, such that the objective function will reduce as the length of the eye openness indicator line increases.

[0122] Although the objective functions and optimisation routines described above relate to minimising a residual function (or loss function), it will be appreciated that the objective function may also be defined such that the optimisation routine determines a maximum value of the objective function. For example, the objective function may be based on a sum of the magnitudes of the angles complementary to α**1** and α**2**—that is a sum of the magnitudes of: an angle between the eye openness indicator line **557** and the first tangent **562**; and an angle between the eye openness indicator line **557** and the second tangent **564**. Such an objective function would have a maximum value of x radians. A maximising objective function may also further include a length term in the sum proportional to the length of the eye openness indicator line **557**.

[0123] The at least one termination condition for the value of the objective function may comprise known termination conditions used in optimisation methods. For example, for residual objective functions (or loss functions) defined for an optimisation routine to minimise the value of the objective function, the at least one termination condition may comprise the value of the objective function being less than a threshold value. For objective functions defined for an optimisation routine to maximise the value of the objective function, the at least one termination condition may comprise the value of the objective function being greater than a threshold value.

[0124] The at least one terminating condition may also relate to a convergence of successive iterative values of the objective function. For example, the at least one terminating condition may comprise successive iterative values of the objective function converging such that a variation of a predetermined number of the successive iterative values is less than a convergence threshold. The convergence threshold may correspond to a relative and/or an absolute variation in a fixed number of successive iterative values (e.g. 5 successive iteration values with an absolute variation of 10^{-6} or with a relative variation of 10^{-4}).

[0125] The at least one terminating condition may comprise reaching a maximum number of iterations of adjusting the values of the first and second intersection points, **558**, **560**.

[0126] The eye openness indicator line **557** in FIG. **4** is not orthogonal to the horizontal axis **540**. This illustrates the limitation of the eye openness measure of FIG. **3** which is limited to eyelid-to-eyelid distances in a direction orthogonal to the horizontal axis **540**. Such a limitation can provide inaccurate or inconsistent values of eye openness for some eyelid edge shapes. In contrast, by defining the eye openness indicator line **557** based on the orthogonality to the two tangents **562**, **564**, the disclosed apparatus and methods can advantageously provide a more accurate representation of eye openness and allow for a greater anatomical variation in eye shape.

[0127] In the example of FIG. **4**, the controller **450** comprises an optional curve fitter **468** and the optimiser **452** receives the first curve data **454** and second curve data **456** via the curve fitter **468**. In this example, the first curve data **454** comprises at least three first data points (also referred to as landmark points or eyelid landmarks) associated with (i.e. lying on, approximately on or proximal to) the first eyelid edge **542**. Similarly, the second curve data **456** comprises at least three second data points associated with the second eyelid edge **544**. The first and second data points may be identified or annotated in the eye image **438** manually by a user or automatically using image processing, as discussed further below.

[0128] The curve fitter **468** may fit a curve to the at least three first data points to define the first curve **553** and fit a curve to the at least three second data points to define the second curve **555**. The curve fitter **468** may fit one or both curves using known curve fitting methods. The curve fitting performed by the curve fitter **468** can advantageously smooth out any annotation errors in the at least three first and second data points. For example, if a user has inaccurately annotated one of the at least three first data points such that it is offset from the first eyelid edge **542**, the curve fitting based on the at least two other first data points can compensate for the annotation error. In some examples, the first and second curve data **454**, **456** respectively comprise three first and second data points. Three data points has surprisingly been found to provide sufficient accuracy and repeatability in the eye openness value **446** for eye tracking applications. Although more data points can provide an improvement in accuracy and/or repeatability, the improvement is at the expense of extra annotation and processing. In some examples, the number of first data points may differ from the number of second data points. For example, different number of data points may be selected when one eyelid is more difficult to annotate or has an atypical anatomical shape.

[0129] The curve fitter **468** may fit one or both curves **553**, **555** using a curve function comprising any of: a second order polynomial function; a polynomial function of an order higher than second order; an elliptical function; a circular function; a parabolic function; a hyperbolic function; a trigonometric function or any other known curve function. Using a second order polynomial function (or other function with a single turning point) can advantageously avoid overfitting of the first and second curves **553**, **555** and the resulting poor representation of the eyelid edge.

[0130] In other examples, the controller **450** may not include a curve fitter and the optimiser **452** may receive the first and second curve data **454**, **456** from an external source, such as another controller or processing module or as user input (manual data entry). The optimiser **452** may receive the first and second curve data **454**, **456** as equation parameters defining the respective curves (for example, parabolic equation parameters or polynomial equation parameters) or as a set of points for defining each respective curve by interpolation. In such examples, the controller **450** may receive the first and second curve data **454**, **456** and may not receive the eye image **438**.

[0131] FIG. **6** schematically illustrates the operational steps performed by a controller comprising the curve fitter **468** and the optimiser **452**. The operation steps include:

[0132] (i) For each of the first (upper) eyelid edge **642** and the second (lower) eyelid edge **644**, receive N landmark points **670**, **672** associated with the respective eyelid edge. N can be any number >2 and different for the first eyelid edge **642** and the second eyelid edge **644**. The N landmark data points for the first eyelid edge **642** comprise at least three first data points **670**. The N landmark data points for the second eyelid edge **642** comprise at least three second data points **672**.

[0133] (ii) Fit 2D first and second curves **653**, **655** (for example a circle segment or a polynomial) in the image **638** to the respective N landmark points **670**, **672**.

[0134] (iii) Perform an optimisation routine as described above to determine the eye openness indicator line **657**. Provide the eye openness value based on the length of the eye openness indicator line.

[0135] Returning to FIG. **4**, the controller **450** can comprise an optional eyelid annotator **466** configured to provide the first and second curve data **454**, **456** to the curve fitter **468**. The eyelid annotator **466** is configured to receive the image of the eye **438**. The eyelid annotator **466** may receive the image of the eye **438** from an image sensor, from a memory storage device or from another processing module which performs initial image processing on the image **438** (e.g. noise filtering etc). The eyelid annotator **466** may include an eyelid landmark detection algorithm configured to identify and output landmark points on the first and second eyelid **542**, **544**. For example, the eyelid annotator **466** may determine the at least three first data points and the at least three second data points using the eyelid landmark detection algorithm.

[0136] In other examples, the controller **450** may not include an eyelid annotator and the curve fitter **468** may receive the first and second curve data **454**, **456** from an external source, such as another controller or processing module or as user input (landmark points based on manually annotated images). In such examples, the controller **450** may receive the first and second curve data **454**, **456** and may not receive the eye image **438**.

[0137] The controller **450** of FIG. **4** can be used in an eye tracking system in which the controller **450** can provide eye openness values for eye images captured by the eye tracking system for applications including blink detection, alertness detection and gaze filtering. The controller **450** can also be used to provide training data for an eye openness ML algorithm.

[0138] FIG. **7** illustrates another controller **750** for providing an eye openness value and an eye tracking system **700** according to embodiments of the present disclosure. Features of FIG. **7** that are also present in FIG. **4** have been given corresponding numbers in the **700** series and are not necessarily described again here.

[0139] The controller **750** is configured to receive a plurality of eye images **738** and first and second curve data **754**, **756** associated with each image. The controller **750** comprises an optimiser **752** which is configured to determine an eye openness indicator line and an eye openness value **746** for each image based on the respective first and second curve data **754**, **756**, by performing the optimisation routine in the same way as described above in relation to FIGS. **4** and **5**.

[0140] The controller **750** is configured to label each eye image in the plurality of eye images with the respective eye openness value **746** to provide labelled training data **774**. The labelled training data **774** may comprise a plurality of data pairs, each data pair comprising an eye image and an eye openness value corresponding to the eye image. The controller **750** is further configured to train an ML algorithm **776** using the labelled training data. For example, the controller **750** may train a convolutional neural network to determine eye openness values of any image of an eye.

[0141] The trained ML algorithm **776** may form part of an eye tracking system **700**. The eye tracking system **700** can receive a further image of an eye and output an eye openness value by processing the further image of the eye using the trained ML algorithm **776**. In some examples, the eye tracking system **700** may comprise the controller **750**.

[0142] FIG. **8** illustrates a method for determining an eye openness value of an image of an eye according to an embodiment of the present disclosure.

[0143] Step **880** comprises receiving first curve data defining a first curve representative of a first eyelid edge in an image of an eye. Step **882** comprises receiving second curve data defining a second curve representative of a second eyelid edge of the image of the eye. Step **884** comprises determining an eye openness indicator line extending from a first intersection point on the first curve to a second intersection point on the second curve, using an optimisation routine. The optimisation routine comprises:

[0144] defining an objective function representative of: an orthogonality of the eye openness indicator line and a tangent to the first curve at the first intersection point; and an orthogonality of the eye openness indicator line and a tangent to the second curve at the second intersection point; and adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied. Step **886** comprises providing an eye openness value based on a length of the eye openness indicator line.

[0145] FIG. **9** illustrates a method of training an ML algorithm according to an embodiment of the present disclosure.

[0146] Step **990** comprises receiving an image of an eye. Step **992** comprises determining an eye openness value according to the method of FIG. **8** and labelling the image with the eye openness value to provide labelled training data. Step **994** comprises training a machine learning algorithm to output eye openness values in response to eye image inputs using the labelled training data. The trained ML algorithm may be used in an eye tracking system to determine an eye openness value for a further image of the eye captured by the eye tracking system.

[0147] The disclosed apparatus and methods can determine eye openness for eye images. The apparatus and methods can infer image domain eye openness from an image of an eye or from a set of landmark points along the upper and lower eyelids of the image. The apparatus and methods can advantageously provide accurate and repeatable values of eye openness for eye images. The eye openness values may be used for labelling training data for an eye openness ML algorithm and/or for determining eye openness values in an eye tracking system.

[0148] The disclosed apparatus and methods can have reduced sensitivity to the placement of landmark points. The apparatus and methods can enable an annotator (machine or human) to be less precise when annotating the eyelid edges or the landmark points on the eyelid edge. For example, a human annotator need only identify three or more points along the eyelid edge. There is no requirement to identify the pair of points on the two eyelid edges with the greatest separation orthogonal to the horizontal axis as required using the measure of FIG. **3**.

[0149] Throughout the present specification, the descriptors relating to relative orientation and position, such as "horizontal", "vertical", "top", "bottom" and "side", are used in the sense of the orientation of the apparatus/device as presented in the drawings. However, such descriptors are not intended to be in any way limiting to an intended use of the described or claimed invention.

[0150] It will be appreciated that any reference to "close to", "before", "shortly before", "after" "shortly after", "higher than", or "lower than", etc, can refer to the parameter in question being less than or greater than a threshold value, or between two threshold values, depending upon the context.

1. A controller configured to:

receive first curve data defining a first curve representative of a first eyelid edge in an image of an eye;

receive second curve data defining a second curve representative of a second eyelid edge in the image of the eye;

determine an eye openness indicator line extending from a first intersection point on the first curve to a second intersection point on the second curve by performing an optimization routine comprising:

defining an objective function representative of:

an orthogonality of the eye openness indicator line to a first tangent to the first curve at the first intersection point; and

an orthogonality of the eye openness indicator line to a second tangent to the second curve at the second intersection point; and

adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied; and

provide an eye openness value based on a length of the eye openness indicator line, wherein the controller is further configured to:

receive the image of the eye and

label the image of the eye with the eye openness value to provide labelled training data.

2. The controller of claim **1**, wherein the objective function is based on:

a first angle between the eye openness indicator line and a normal to the tangent to the first curve at the first intersection point; and

a second angle between the eye openness indicator line and a normal to the tangent to the second curve at the second intersection point.

3. The controller of claim **1**, wherein the objective function is based on:

a dot product of the eye openness indicator line and the tangent to the first curve at the first intersection point; and

a dot product of the eye openness indicator line and the tangent to the second curve at the second intersection point.

4. The controller of claim **3**, wherein the objective function is based on a sum of squares of, or a sum of magnitudes of:

the dot product of the eye openness indicator line and the tangent to the first curve at the first intersection point; and

the dot product of the eye openness indicator line and the tangent to the second curve at the second intersection point.

5. The controller of claim **1**, wherein the optimization routine comprises adjusting the value of the first intersection point and the value of the second intersection point to reduce a value of the objective function until the at least one termination condition for the value of the objective function is satisfied.

6. The controller of claim **1**, wherein the at least one termination condition comprises at least one of:

the value of the objective function is less than a threshold value;

the value of the objective function is greater than a threshold value;

a variation of a fixed number of successive values of the objective function converge within a convergence threshold; and

a maximum number of iterations of adjusting the value of the first intersection point and the value of the second intersection point have been performed.

7. The controller of claim **1**, wherein the controller is configured to:

receive the first curve data as first curve data comprising at least three first data points associated with the first eyelid edge;

receive the second curve data as second curve data comprising at least three second data points associated with the second eyelid edge;

curve fit the at least three first data points to define the first curve; and

curve fit the at least three second data points to define the first curve.

8. The controller of claim **7**, wherein the controller is configured to curve fit the at least three first data points and the at least three second data points using any of:

a second order polynomial function;

a polynomial function of an order higher than second order; an elliptical function;

a parabolic function;

a hyperbolic function; and

a trigonometric function.

9. The controller system of claim **7**, wherein the controller is configured to:

receive the image of the eye;

determine the at least three first data points using image processing; and

determine the at least three second data points using image processing.

10. The controller of claim **1** configured to:

train a machine learning algorithm to output eye openness values in response to eye image inputs using the labelled training data.

11. A machine learning algorithm trained by the controller of claim **10**.

12. An eye tracking system comprising:

a memory storing the machine learning algorithm of claim **11**; and

a processor configured to:

receive a further image of an eye; and

output an eye openness value by processing the image of the eye using the machine learning algorithm.

13. An eye tracking system comprising the controller of claim **1**.

14. A method for determining an eye openness value of an image of an eye, the method comprising:

receiving first curve data defining a first curve representative of a first eyelid edge in an image of an eye;

receiving second curve data defining a second curve representative of a second eyelid edge in the image of the eye;

determining an eye openness indicator line extending from a first intersection point on the first curve to a second intersection point on the second curve by performing an optimization routine comprising:

defining an objective function representative of:

an orthogonality of the eye openness indicator line and a tangent to the first curve at the first intersection point; and

an orthogonality of the eye openness indicator line and a tangent to the second curve at the second intersection point; and

adjusting a value of the first intersection point and a value of the second intersection point until at least one termination condition for a value of the objective function is satisfied;

providing an eye openness value based on a length of the eye openness indicator line;

labelling a received image of the eye with the eye openness value to provide labelled training data.

15. The method of claim **14** further comprising:

receiving the image of the eye;

labelling the image of the eye with the eye openness value to provide labelled training data; and

training a machine learning algorithm to output eye openness values in response to eye image inputs using the labelled training data.

16. A machine learning algorithm obtainable by the method of claim **15**.

17. An eye tracking system comprising:

a memory storing the machine learning algorithm of claim **16**; and

a processor configured to:

receive a further image of an eye; and

output an eye openness value by processing the image of the eye using the machine learning algorithm.

18. A head-mounted device comprising the eye tracking system of claim **12**.

19. One or more non-transitory computer-readable storage media storing computer-executable instructions that, when executed by a computing system, causes the computing system to perform the method of claim **14**.

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